

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Probability & Statistics**

Code No: Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

11. Differentiate between Convenient sampling and Systematic random sampling with suitable examples.
12. From the following data, find Mean and Median. Then use an empirical relation across mean, median and mode to find Mode.
13. When Mean is 34.69 for 65 total observations, find the missing frequency from the following data.
14. Use Karl Pearson's correlation method to examine the relationship between marks obtained in two subjects:
15. The number of defective articles is found to be 2% in a lot of 300 sample articles per day, then find out following probabilities using suitable probability distribution that(i) There are exactly two defective articles per day(ii) There are at least 2 defective articles per day.
16. Six coins are tossed 500 times such that probability of getting a head is 21% in each toss. Fit a Binomial distribution for a number of heads with the following frequency distribution:
17. In a Hospital, the height of patients follows normal distribution with mean height of 53 inches with standard deviation of 3 inches. Then find out the following probabilities of heights of patients that: i) less than 57 inches, ii) more than 64 inches, iii) between 50 and 56 inches. Use Normal table.

Group C

Attempt any TWO questions [2x10=20]

18. Automobile Company Sells Cars of different model during a year. Data are presented in the following table:
19. An analysis of daily wages, paid to workers in two firms A and B belonging to the same industry, gives the following results:
20. Perform a one-way ANOVA to differentiate average sales of computers from three electronic shops as given below at 5% level of significance: